Ch 4.3

1. $a_1 = (1,1,1,1)$, $a_2 = (0,1,3,4)$

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

$$A^TA = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}$$

$$A^Tb = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$A^TA\hat{x} = A^Tb \Rightarrow \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \hat{x} = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 8 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 36 \\ 40 \end{bmatrix} \qquad 10x_2 = 40$$
$$x_2 = 4$$

$$4x_1 + 32 = 36 \quad \therefore 4x_1 = 4 \quad \therefore x_1 = 1$$

$$\therefore \hat{x} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 13 \\ 17 \end{bmatrix}$$

$$e = A\hat{x} = \begin{bmatrix} 1 \\ 5 \\ 13 \\ 17 \end{bmatrix}$$

$e = b - p = \begin{bmatrix} -1 \\ 3 \\ -5 \\ 3 \end{bmatrix}$

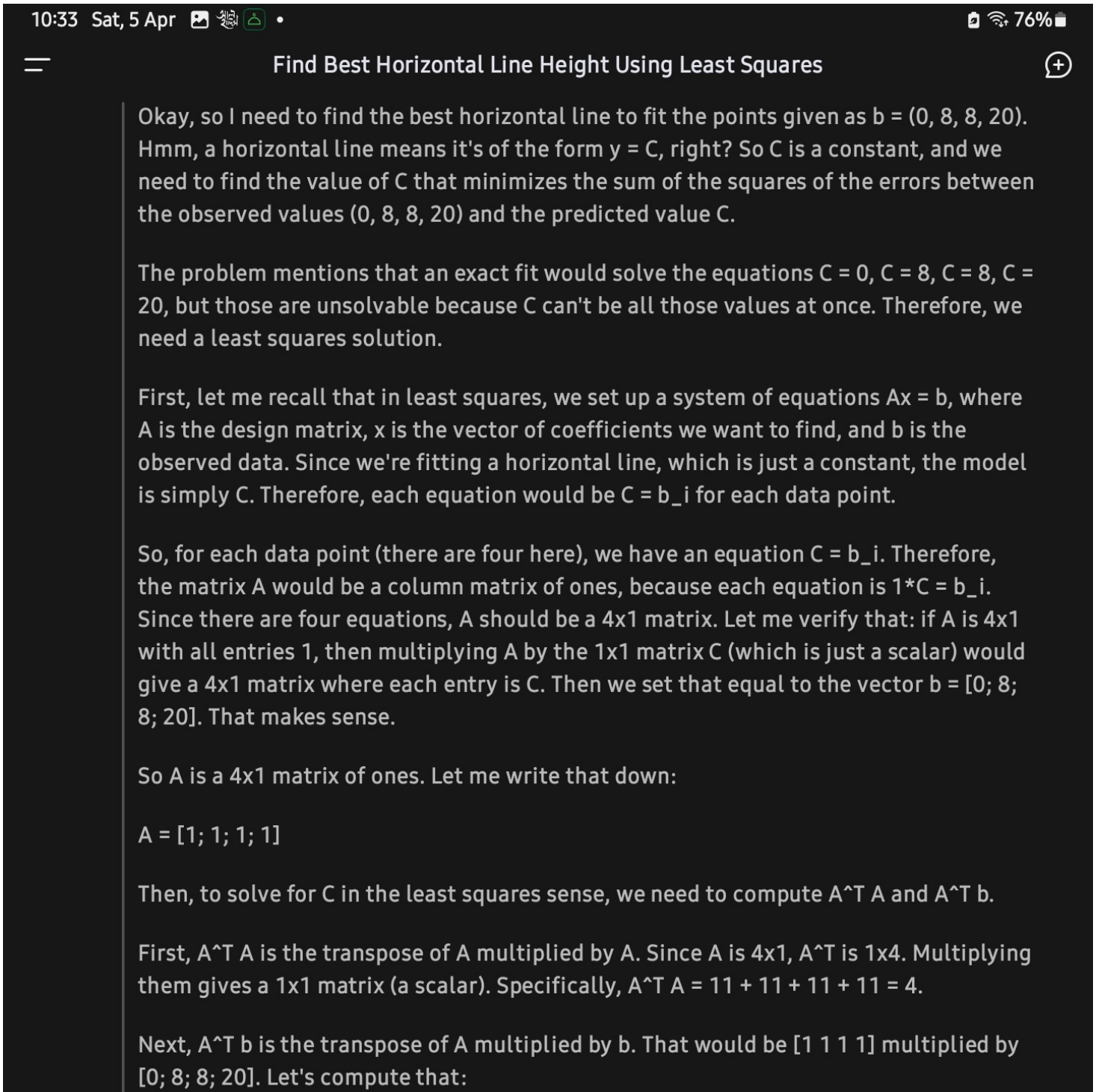
$E = \|e\|^2 = 44$

$1 + 9 + 9 + 25$

$10 + 34$

4

5



Okay, so I need to find the best horizontal line to fit the points given as b = (0, 8, 8, 20). Hmm, a horizontal line means it's of the form y = C, right? So C is a constant, and we need to find the value of C that minimizes the sum of the squares of the errors between the observed values (0, 8, 8, 20) and the predicted value C.

The problem mentions that an exact fit would solve the equations C = 0, C = 8, C = 8, C = 20, but those are unsolvable because C can't be all those values at once. Therefore, we need a least squares solution.

First, let me recall that in least squares, we set up a system of equations Ax = b, where A is the design matrix, x is the vector of coefficients we want to find, and b is the observed data. Since we're fitting a horizontal line, which is just a constant, the model is simply C. Therefore, each equation would be C = b_i for each data point.

So, for each data point (there are four here), we have an equation C = b_i. Therefore, the matrix A would be a column matrix of ones, because each equation is 1*C = b_i. Since there are four equations, A should be a 4x1 matrix. Let me verify that: if A is 4x1 with all entries 1, then multiplying A by the 1x1 matrix C (which is just a scalar) would give a 4x1 matrix where each entry is C. Then we set that equal to the vector b = [0; 8; 8; 20]. That makes sense.

So A is a 4x1 matrix of ones. Let me write that down:

A = [1; 1; 1; 1]

Then, to solve for C in the least squares sense, we need to compute A^T A and A^T b.

First, A^T A is the transpose of A multiplied by A. Since A is 4x1, A^T is 1x4. Multiplying them gives a 1x1 matrix (a scalar). Specifically, A^T A = 11 + 11 + 11 + 11 = 4.

Next, A^T b is the transpose of A multiplied by b. That would be [1 1 1 1] multiplied by [0; 8; 8; 20]. Let's compute that:

10 + 18 + 18 + 120 = 0 + 8 + 8 + 20 = 36.

So, the normal equation A^T A C = A^T b becomes 4C = 36. Solving for C gives C = 36 / 4 = 9.

Therefore, the best horizontal line is y = 9.

But let me check if I did everything right. Maybe I made a mistake in setting up the matrix A? Let's see. If we have four data points, each with a single feature (which is just 1, since it's a constant term), then A should indeed be a column of ones. The equations are C = 0, C = 8, etc., so each row of A is 1, and the coefficient is C. So yeah, that seems correct.

Alternatively, maybe I should think of this as a linear regression problem with no independent variables, just the intercept. In that case, the intercept is indeed the mean of all the observations. So the least squares estimate for C is the average of the b values. Let's check: (0 + 8 + 8 + 20)/4 = (36)/4 = 9. Yep, that's the same result.

So that checks out. Therefore, C is 9.

Now, the problem also mentions drawing the horizontal line at height £ = C and the four errors in e. I assume £ is a typo and should be C. The errors e would be the differences between each observed value and the predicted value C. So for each data point:

e_1 = 0 - 9 = -9

e_2 = 8 - 9 = -1

e_3 = 8 - 9 = -1

e_4 = 20 - 9 = 11

So the error vector e is [-9, -1, -1, 11]^T.

Let me also confirm that these errors are orthogonal to the column space of A. Since the column space of A is spanned by the vector [1,1,1,1]^T, the dot product of e and A should be zero. Let's compute that:

e • A = (-9)(1) + (-1)(1) + (-1)(1) + 11)(1) = -9 -1 -1 +11 = 0.

Yes, that's zero, which confirms that the error is orthogonal to the column space, as expected in least squares.

So, to recap, the matrix A is a 4x1 column of ones. The normal equation A^T A C = A^T b gives 4C = 36, leading to C = 9. The errors are computed as the difference between each b_i and C, resulting in the vector e = [-9, -1, -1, 11]^T.

Therefore, the height C of the best horizontal line is 9, and the errors are as calculated. Drawing this would involve plotting the horizontal line at y=9 and then vertical lines (or arrows) from each data point to the line, representing the errors. For the points (0,0), (1,8), (2,8), (3,20) connected to their projections (0,9), (1,9), (2,9), (3,9) respectively. The lengths of these vertical lines are the absolute values of the errors: 9, 1, 1, 11 units respectively.

I think that's all. Let me make sure there are no other steps. The problem specifically asks to find the matrix A, solve A^T A Z = A^T b (where Z is C in this case), and then mention drawing the line and errors. I've addressed each part. The key was recognizing that a horizontal line fit is equivalent to finding the mean of the y-values, which is the least squares solution here.

Another way to think about it is that minimizing the sum of squared errors Σ(b_i - C)^2 leads to taking the derivative with respect to C, setting it to zero. The derivative would be 2Σ(b_i - C)(-1) = 0 => Σ(b_i - C) = 0 => Σb_i = 4C => C = Σb_i /4, which is the mean. So that's another confirmation.

Therefore, everything seems consistent. I think I've got it right.

Final Answer
The height of the best horizontal is boxed{9}.

New chat

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DeepThink (R1) Search

10:33 Sat, 5 Apr • 76%

Find Best Horizontal Line Height Using Least Squares

Final Answer
The height of the best horizontal line is\boxed{9}.

To find the height C of the best horizontal line to fit the points $b = (0, 8, 8, 20)$, we need to solve the problem using the least squares method.

1. Set up the matrix A:
   - Since we are fitting a horizontal line $y = C$, the matrix A is a column of ones because each equation is $C = b_i$.
   - Therefore, A is a 4x1 matrix:

$$A = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

2. Form the normal equations:
   - Compute $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = 4$$

   - Compute $A^T b$:

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} = 0 + 8 + 8 + 20 = 36$$

3. Solve the normal equation:
   - The equation $A^T A Z = A^T b$ becomes $4Z = 36$.
   - Solving for Z (which is C):

- Solving for C (which is $\bar{C}$):

$$C = \frac{36}{4} = 9$$

4. Calculate the errors:
   - The errors e are the differences between each observed value and the predicted value C:

$$e = \begin{bmatrix} 0-9 \\ 8-9 \\ 8-9 \\ 20-9 \end{bmatrix} = \begin{bmatrix} -9 \\ -1 \\ -1 \\ 11 \end{bmatrix}$$

The height of the best horizontal line is $\boxed{9}$. The errors are $[-9, -1, -1, 11]^T$, which can be visualized as vertical distances from each data point to the line $y = 9$.

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DeepThink (R1) Search

A → cols of 1 cz scaling the numbers C equally for each row → horizontal line.

↳ every o/p is C. Same val repeated

tryna fit $y = c$

→ No slope

→ a st line across the graph at height C.

→ every x gives same y.

∴ $C = 9$ → (1, 9), (2, 9), (3, 9) ...

Given: $b = [0, 8, 8, 20]$

tryna find 1 num c that's close to all these nums. → fit a hz line to data.

y always same

Now, we solve

$Ax = b$

↓ ↓ ↓

model mat | c | data

to make each row say: $C = bi$

$$c = \frac{0+8+8+20}{4} = 9$$

(6) → [0]

$\begin{bmatrix} 9 \\ 9 \\ 9 \\ 9 \end{bmatrix}$

Project b = (0, 8, 8, 20) onto the line througha = (1,1,1,1). Find $\hat{x}$ = a'b/aTa and the projection p = $\hat{x}$a. Check that e = b — p is perpendicular to a, and find the shortest distance ||e|| from b to the line through a.

$e^T a = (-9, -1, -1, 11)^T (1, 1, 1, 1)$

$= 0$

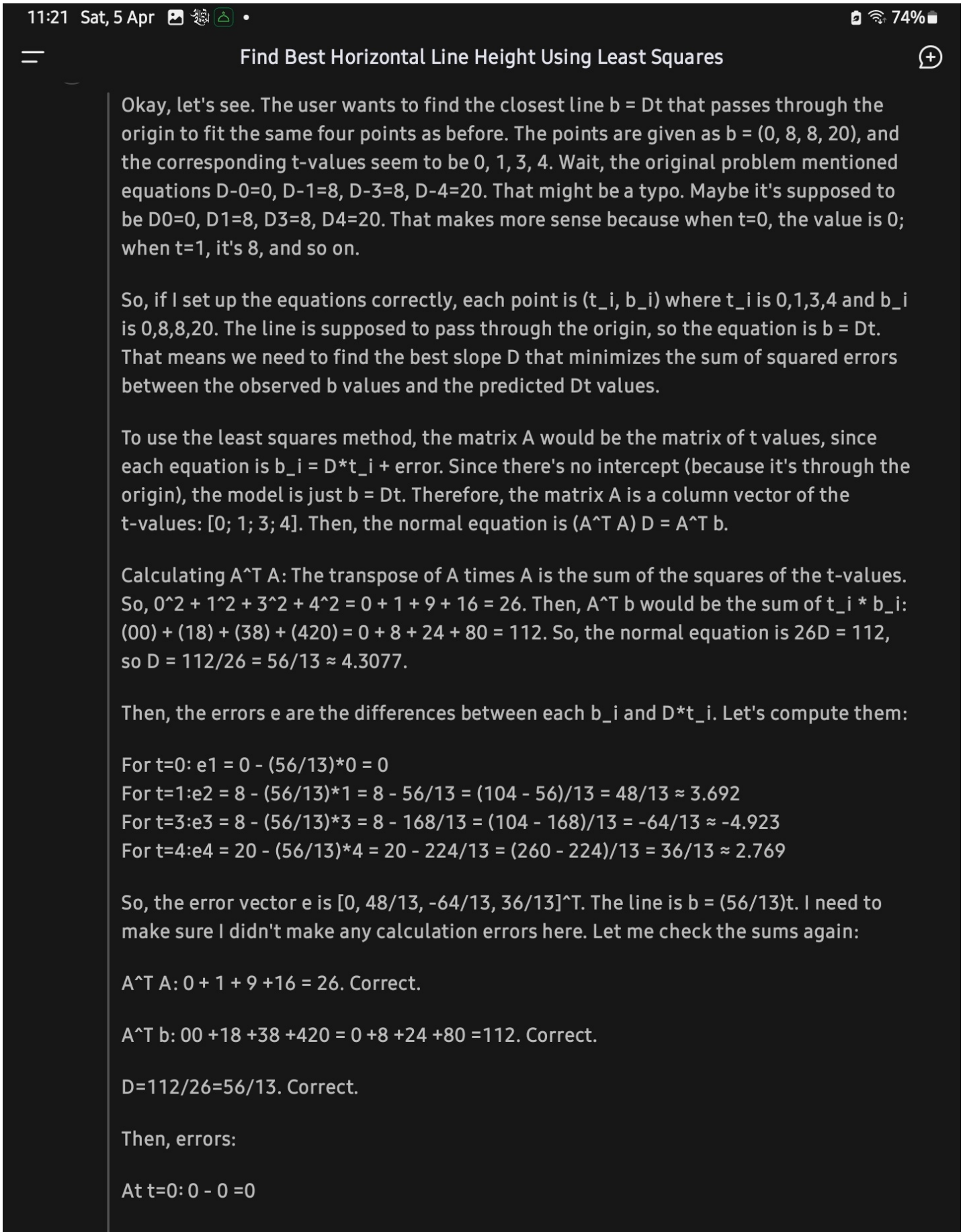
$\|e\| = \sqrt{204}$

$81 + 2 + 121$

→ shortest distance from b to the line thru a.

Squared error = "How wrong are we, in total?"

Shortest distance = "How far is the point from the line directly?"



t=1: 8 -56/13= (104-56)/13=48/13

t=3:8 -168/13= (104-168)/13= -64/13

t=4:20 -224/13= (260-224)/13=36/13

Yes, that's right.

So the best line is D=56/13 and the errors are as calculated. The figure would show the line passing through the origin with slope 56/13, and vertical errors at each data point. The points are (0,0), (1,8), (3,8), (4,20). The line passes through (0,0), and at t=1, the predicted value is 56/13 ≈4.307, so the error is 8 -4.307≈3.693. Similarly for others. Drawing those vertical distances would show the errors.

To find the closest line $b = Dt$ through the origin to the points \(t, b) = (0, 0), (1, 8), (3, 8), (4, 20)\), we use the least squares method:

1. Matrix Setup:
   - The system of equations is $Dt_i = b_i$, leading to the matrix A (column of t -values):

$$A = \begin{bmatrix} 0 \\ 1 \\ 3 \\ 4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}.$$

2. Normal Equations:
   - Compute $A^T A$:

$$A^T A = 0^2 + 1^2 + 3^2 + 4^2 = 26.$$

   - Compute $A^T b$:

$$A^T b = (0 \cdot 0) + (1 \cdot 8) + (3 \cdot 8) + (4 \cdot 20) = 0 + 8 + 24 + 80 = 112.$$

3. Solve for D:
   - The equation $(A^T A)D = A^T b$ $D = 112$, so:

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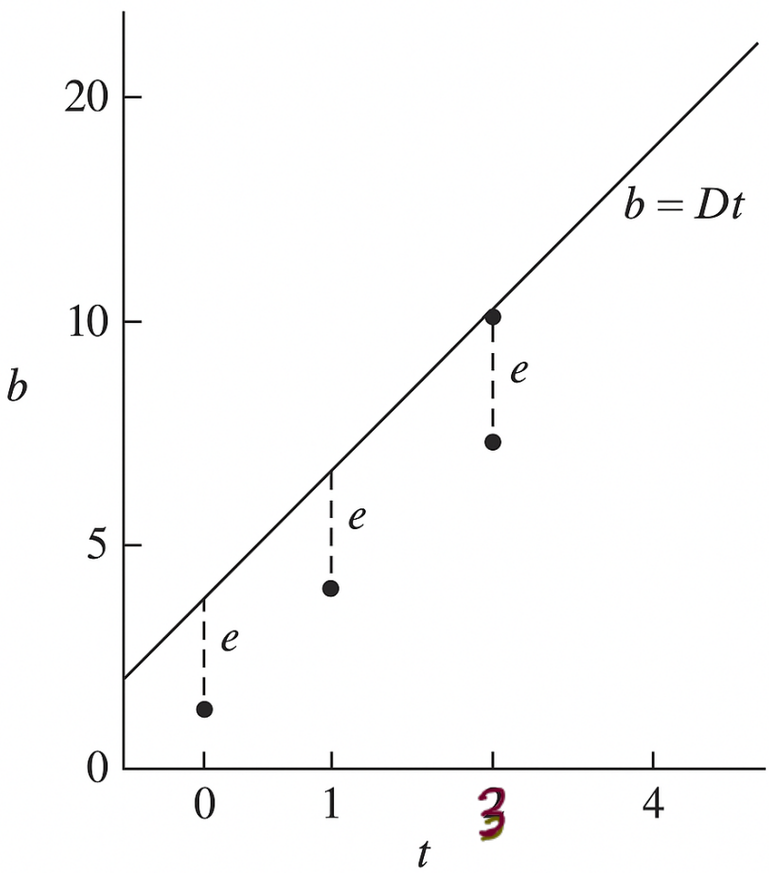


$$b = C + Dt + Et^2$$

For the points $t_1 = 0$, $t_2 = 1$, $t_3 = 3$, and $t_4 = 4$, the system becomes:

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}$$

Step 2: Setting Up the Normal Equations

Next, let's form the normal equations $A^T Az = A^T b$. First, we compute $A^T A$ and $A^T b$.

Compute $A^T A$:

$$A^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 9 & 16 \end{bmatrix}$$

Now, calculate $A^T A$:

$$A^T A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 9 & 16 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix}$$

Compute $A^T b$:

Now, compute $A^T b$ using the new vector $b = \begin{bmatrix} 0 & 8 & 8 & 20 \end{bmatrix}$:

$$A^T b = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 4 \\ 0 & 1 & 9 & 16 \end{bmatrix} \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}$$

Step 3: The Normal Equations

The updated normal equations become:

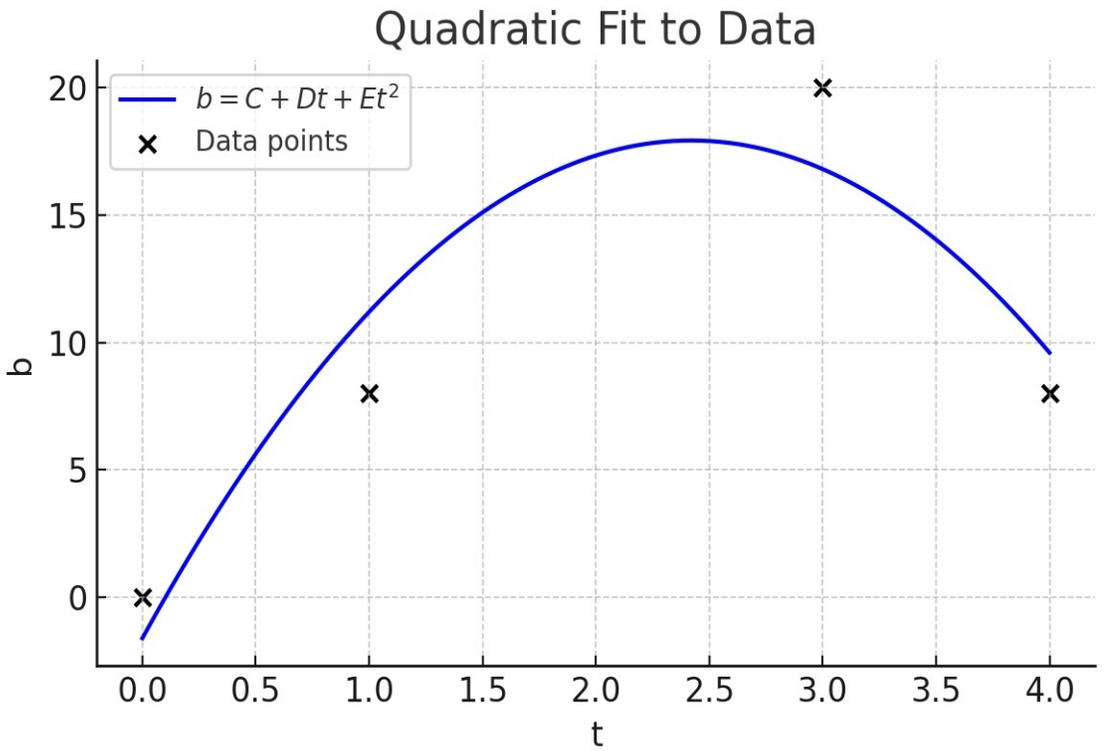
$$A^T Az = A^T b$$

$$\begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix} \begin{bmatrix} C \\ D \\ E \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}$$

[05/04, 12:27] Shara: import numpy as np
import matplotlib.pyplot as plt

```
# Data points (t, b)
t = np.array([0, 1, 3, 4])
b = np.array([0, 8, 20, 8])

# Create design matrix A for quadratic model: b = C + D*t + E*t^2
A = np.vstack((np.ones_like(t), t, t**2)).T

# Solve for coefficients using least squares: x = (A^T A)^(-1) A^T b
ATA = A.T @ A
ATb = A.T @ b
x = np.linalg.solve(ATA, ATb)

# Evaluate fitted curve
t_fit = np.linspace(min(t), max(t), 100)
b_fit = x[0] + x[1]*t_fit + x[2]*t_fit**2

# Plot
plt.figure(figsize=(6, 4))
plt.plot(t_fit, b_fit, label=r'$b = C + Dt + Et^2$', color='blue')
plt.scatter(t, b, color='black', label='Data points')
plt.title('Quadratic Fit to Data')
plt.xlabel('t')
plt.ylabel('b')
plt.legend()
plt.grid(True)
```

plt.tight_layout()
plt.show(), x, ATA, ATb
[05/04, 12:28] Shara: Result
(None,
 array([-1.6   , 16.13333333, -3.33333333]),
 array([[4,   8, 26],
        [8, 26, 92],
        [26, 92, 338]]),
 array([36, 100, 316]))

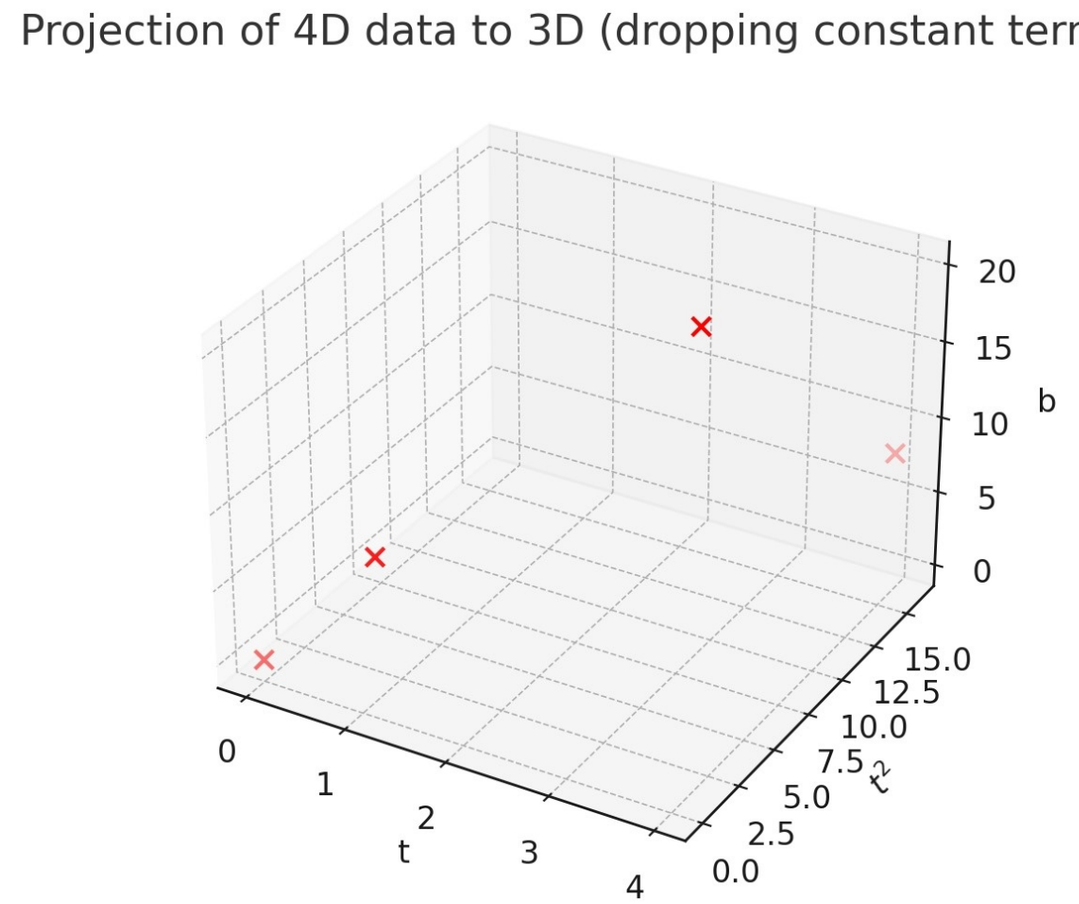


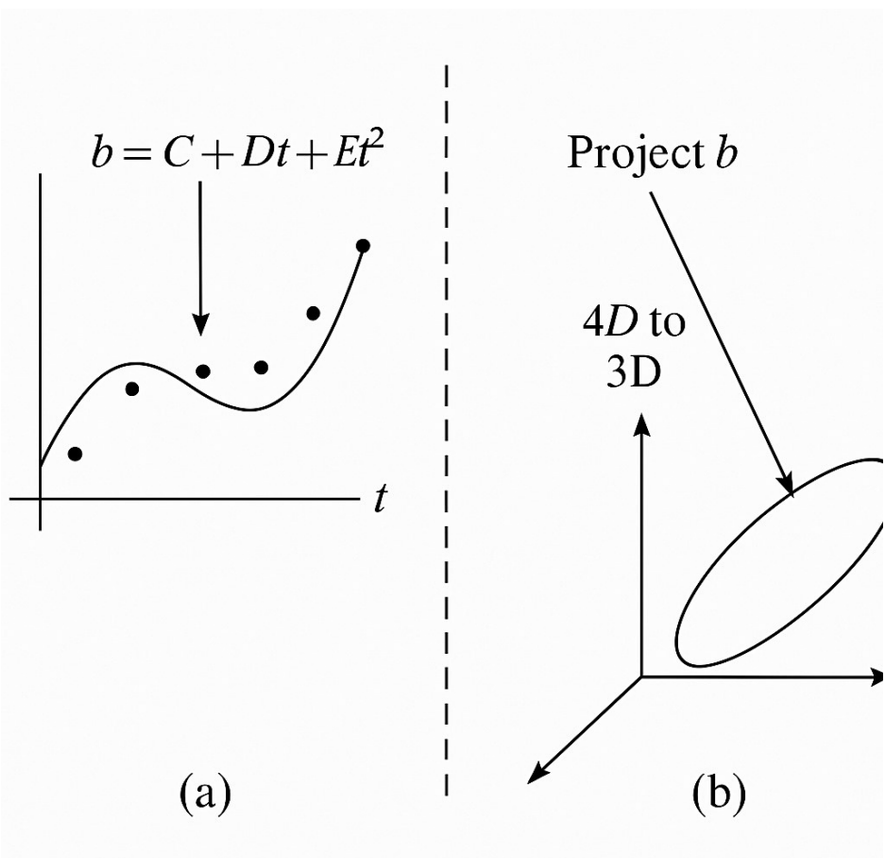
from mpl_toolkits.mplot3d import Axes3D

# Project 4D data (t, t^2, 1, b) into 3D by ignoring one component, e.g., drop '1' (constant term)
# 4D vector: [1, t, t^2, b] → Drop '1' → 3D: [t, t^2, b]
projected_points = np.vstack((t, t**2, b)).T

# 3D scatter plot
fig = plt.figure(figsize=(6, 5))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(projected_points[:, 0], projected_points[:, 1], projected_points[:, 2], c='red', s=50)

```
ax.set_xlabel('t')
ax.set_ylabel(r'$t^2$')
ax.set_zlabel('b')
ax.set_title('Projection of 4D data to 3D (dropping constant term)')

plt.tight_layout()
plt.show()
```





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[05/04, 12:36] Shara: # Updated b vector as per correction: [0, 8, 8, 20]
b_new = np.array([0, 8, 8, 20])

# Construct Vandermonde matrix A for t = [0, 1, 3, 4]
t_vals = np.array([0, 1, 3, 4])
A_cubic = np.vstack((np.ones_like(t_vals), t_vals, t_vals**2, t_vals**3)).T

# Solve A z = b exactly (no least squares needed)
z = np.linalg.solve(A_cubic, b_new)

# Evaluate fitted cubic at dense points for plotting
t_dense = np.linspace(min(t_vals), max(t_vals), 200)
b_cubic = z[0] + z[1]*t_dense + z[2]*t_dense**2 + z[3]*t_dense**3

# Plot
plt.figure(figsize=(6, 4))
plt.plot(t_dense, b_cubic, label='Cubic fit', color='green')
plt.scatter(t_vals, b_new, color='black', label='Data points')
plt.title('Exact Cubic Interpolation')
plt.xlabel('t')
plt.ylabel('b')
plt.grid(True)
plt.legend()
plt.tight_layout()

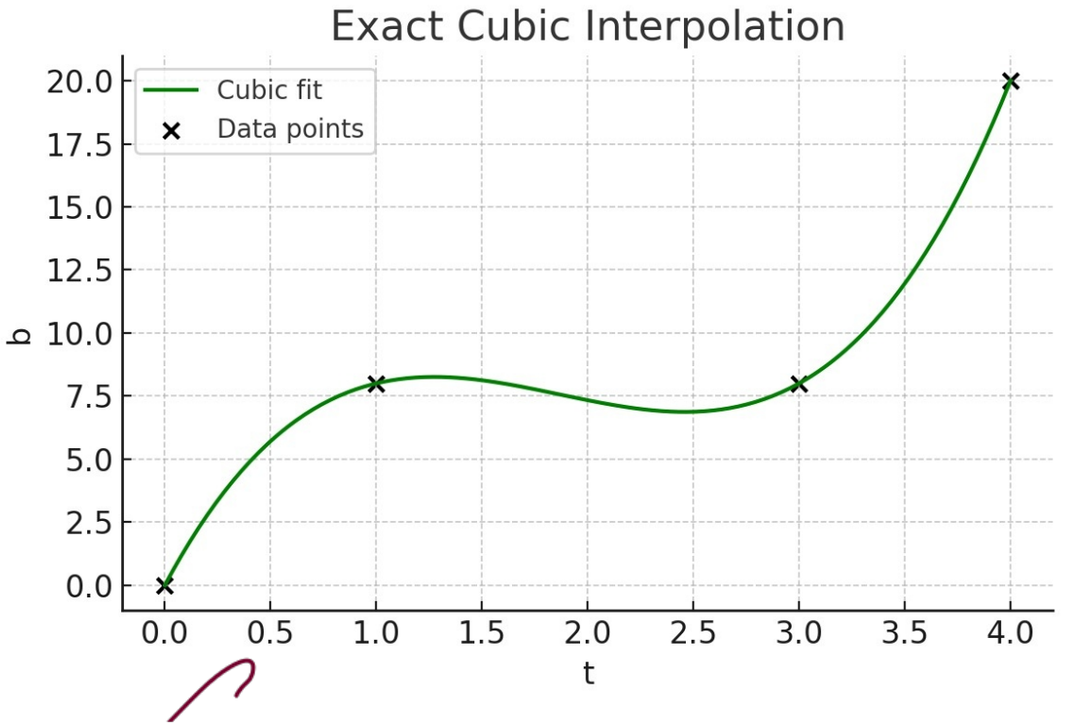
plt.show(), z

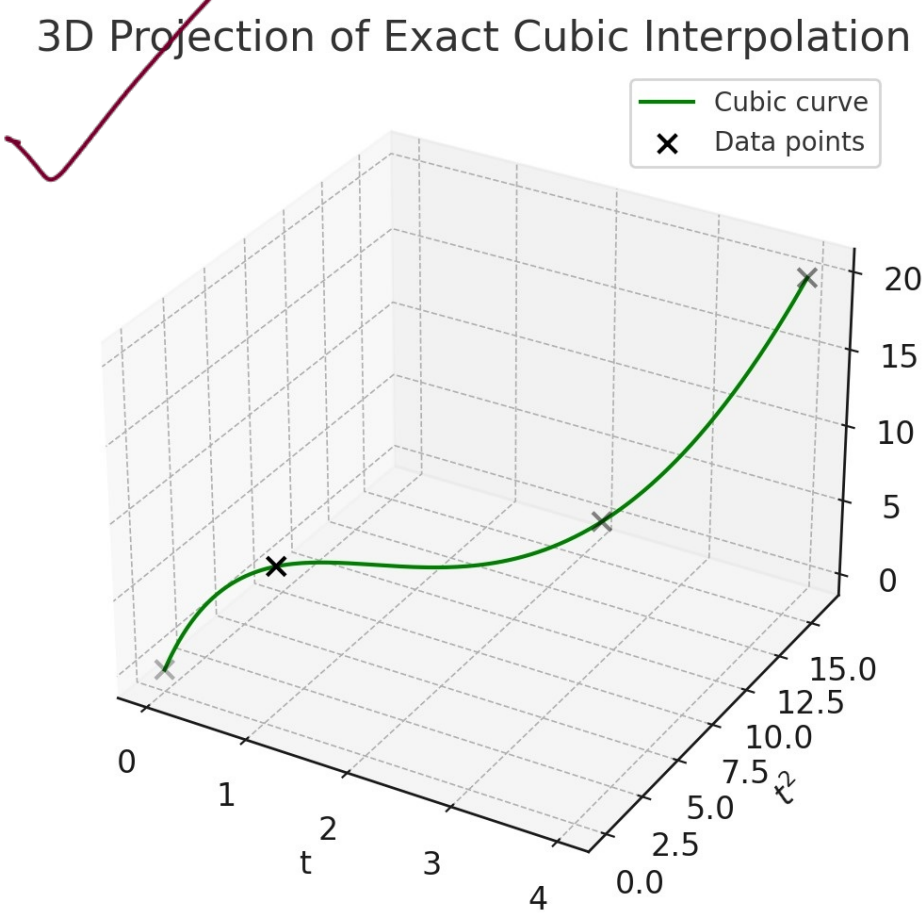
[05/04, 12:37] Shara: # Prepare 3D points for plotting: (t, t^2, b)
projected_3d_points = np.vstack((t_vals, t_vals**2, b_new)).T
cubic_3d_curve = np.vstack((t_dense, t_dense**2, b_cubic)).T

# 3D plot of cubic interpolation
fig = plt.figure(figsize=(7, 5))
ax = fig.add_subplot(111, projection='3d')

# Plot curve and data points
ax.plot(cubic_3d_curve[:, 0], cubic_3d_curve[:, 1], cubic_3d_curve[:, 2], color='green', label='Cubic curve')
ax.scatter(projected_3d_points[:, 0], projected_3d_points[:, 1], projected_3d_points[:, 2], color='black', s=50, label='Data points')

ax.set_xlabel('t')
ax.set_ylabel(r't^2')
ax.set_zlabel('b')
ax.set_title('3D Projection of Exact Cubic Interpolation')
ax.legend()
plt.tight_layout()
plt.show()





12:45 Sat, 5 Apr • 70%

Find Best Horizontal Line Height Using Least Squares

1. Quadratic Fit (Parabola):

Goal: Fit 4 data points \(t, b) = (0, 0), (1, 8), (3, 8), (4, 20)\) with a quadratic model \(b = C + Dt + Et^2 \).

Matrix Setup:

- Design Matrix A (4x3):

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix}$$

  Columns correspond to $1, t, t^2$.

- Vector b (observed values):

$$b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}.$$

Normal Equations:

The system $A\mathbf{x} = b$ is overdetermined (4 equations, 3 unknowns).
Solve via least squares:

$$A^T A\mathbf{x} = A^T b,$$

where:

- A^TA (3x3):

$$\begin{bmatrix} 4 & 8 & 26 \\ 8 & 26 & 92 \\ 26 & 92 & 338 \end{bmatrix}$$

- A^Tb:

$$\begin{bmatrix} 36 \\ 112 \\ 400 \end{bmatrix}.$$

Solving $A^TA\mathbf{x} = A^Tb$ gives coefficients C, D, E.

Interpretation:

- Projection: The quadratic fit projects b from $\mathbb{R}^4$ (4D space of data points) onto a 3D subspace spanned by $1, t, t^2$.
- Residuals: The errors $e = b - A\mathbf{x}$ are minimized in the least squares sense.

---

2. Cubic Fit (Exact Interpolation):

Goal: Fit the same 4 points with a cubic model $b = C + Dt + Et^2 + Ft^3$.

Vandermonde Matrix:

- Design Matrix A (4x4):

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 3 & 9 & 27 \\ 1 & 4 & 16 & 64 \end{bmatrix}$$

  Columns correspond to $1, t, t^2, t^3$.

Exact Solution:

- The system $A\mathbf{x} = b$ is square (4 equations, 4 unknowns).
- Solve directly (no least squares needed):

$$\mathbf{x} = A^{-1}b = \frac{1}{3}\begin{bmatrix} 0 \\ 47 \\ -28 \\ 5 \end{bmatrix}.$$

- Result: The cubic polynomial passes through all 4 points exactly (\(e = 0 \).

Key Differences:

Quadratic Fit	Cubic Fit
Approximate solution(least squares).	Exact solution(interpolation).
Projects b onto a 3D subspace.	Uses full 4D space(no projection).

Errors $e \neq 0$.	Errors $e = 0$.
Overdetermined system.	Square system.

Visualization:

- Figure 4.9a: Shows the quadratic curve $b = C + Dt + Et^2$ approximating the 4 points, with vertical residuals e.
- Figure 4.9b: Represents the projection in $\mathbb{R}^4$:
  - Quadratic model: b projected onto a 3D subspace.
  - Cubic model: b lies in the 4D space (no projection needed).

Final Answer:

$$\boxed{\begin{array}{l}\text{Quadratic fit : Project } b \text{ from 4D to 3D using least squares.} \\ \text{Cubic fit : Exact interpolation in 4D using a Vandermonde matrix.}\end{array}}$$

New chat

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DeepThink (R1) Search

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12:48 Sat, 5 Apr • 69%

Find Best Horizontal Line Height Using Least Squares

Okay, so I need to tackle part (a) and (b) here. Let me start with part (a). The user is saying that the average of the four times t is (0 + 1 + 3 + 4)/4 = 2, and the average of the four b's is (0 + 8 + 8 + 20)/4 = 9. So the center point is (2, 9). They want me to verify that the best line goes through this center point.

First, I remember that when you do a linear regression, the best fit line should pass through the mean of the x and y values. That's a key property of the least squares line. So in this case, since the average t is 2 and the average b is 9, the point (2, 9) should lie on the best fit line.

But wait, the user also mentioned that the best line is x = 1 + 4t. Let me check if that line passes through (2,9). If I plug t=2 into the line equation: x = 1 + 4*2 = 1 + 8 = 9. So yes, when t=2, x=9, which matches the center point (2,9). That seems to verify part (a).

Now, part (b) asks why $C + D\bar{t} = \bar{b}$ comes from the first equation in $A^TA\hat{x} = A^Tb$. Let me

recall the normal equations. The matrix A for linear regression would have a column of ones for the intercept C and a column of t values for the slope D. So A is a 4x2 matrix. Then A^TA is a 2x2 matrix, and A^Tb is a 2x1 vector.

The first equation in $A^TA\hat{z} = A^Tb$ would correspond to the equation for the intercept C. If I compute A^TA, the first row would be the sum of the ones (which is the number of data points, m=4) and the sum of the t values. The first element of A^Tb would be the sum of the b values. So the first equation would be: $mC + (\Sigma t)D = \Sigma b$. If we divide both sides by m, we get $C + D(\Sigma t/m) = \Sigma b/m$. That's exactly $C + D\bar{t} = \bar{b}$. So that explains why the first equation leads to the line passing through the mean point $(\bar{t}, \bar{b})$.

Wait, the user also mentions that in part (a), the best line is given as x = 1 + 4t. Let me check if that's correct. Earlier, when verifying, plugging t=2 gave x=9, which matches the center point. But where did they get 1 and 4 from? Let's do the normal equations properly.

The normal equations for the line C + Dt are:

$\Sigma b = mC + D\Sigma t$
$\Sigma(tb) = C\Sigma t + D*\Sigma t^2$

Given the data points: t = [0,1,3,4], b = [0,8,8,20]

So $\Sigma t = 0+1+3+4=8$
$\Sigma b= 0+8+8+20=36$
$\Sigma t^2= 0^2 +1^2 +3^2 +4^2=0+1+9+16=26$
$\Sigma(tb) = 00+18 +38 +4*20=0+8+24+80=112$

So the normal equations are:

$4C + 8D = 36$
$8C+26D =112$

Solving the first equation: $4C = 36 -8D \rightarrow C = 9 -2D$

Substitute into the second equation: $8*(9 -2D) +26D =112$
$72-16D +26D =112$
$72+10D =112$
$10D=40 \rightarrow D=4$

Then $C=9 -24=1$. So the line is $C + Dt =1 +4t$, which matches the user's statement. So when t=2, $C +42=9$, which is the center point. So everything checks out.

For part (b), the first equation in $A^TA\hat{z} = A^Tb$ is indeed $mC + D\Sigma t = \Sigma b$. Dividing by m gives $C + D*(\Sigma t/m) = \Sigma b/m \rightarrow C + D*\bar{t} = \bar{b}$. This shows that the best fit line must pass through the average point $(\bar{t}, \bar{b})$. That's why the line goes through (2,9) in part (a).

So, putting it all together, part (a) is verified by plugging the average t into the best fit line equation and confirming it gives the average b. Part (b) explains this result by showing that the first normal equation directly leads to the line passing through the mean point.

Step-by-Step Explanation and Verification

**Part (a): Verify the best line passes through the center point \(\bar{t},

\bar{b}) = (2, 9)\)**

1. Data:
   - Times $t = [0, 1, 3, 4]$
   - Values $b = [0, 8, 8, 20]$
   - Averages: $\bar{t} = \frac{0+1+3+4}{4} = 2$, $\bar{b} = \frac{0+8+8+20}{4} = 9$.
2. Best Fit Line:
   - Given as $b = 1 + 4t$ (from solving the normal equations below).
3. Verification:
   - Substitute $\bar{t} = 2$ into the line: $b = 1 + 4(2) = 9$.
   - Result: The point \(2, 9)\) lies on the line \(b = 1 + 4t\). Verified.

Part (b): Explain why $C + D\bar{t} = \bar{b}$ comes from the first equation in $A^T A\hat{x} = A^T b$

1. Normal Equations:
   - For the linear model $b = C + Dt$, the design matrix A is:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix}.$$

   - Compute $A^T A$ and $A^T b$:

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}.$$

   - First Normal Equation:

$$4C + 8D = 36.$$

2. Derivation of $C + D\bar{t} = \bar{b}$:
   - Divide the first equation by $m = 4$:

$$C + D\left(\frac{8}{4}\right) = \frac{36}{4} \quad \Rightarrow \quad C + D\bar{t} = \bar{b}.$$

   - Interpretation: The best line must pass through the centroid \(\bar{t}, \bar{b})\).

Key Results

- Best Fit Line: $b = 1 + 4t$ passes through \(2, 9)\).
- Centroid Condition: $C + D\bar{t} = \bar{b}$ is enforced by the first normal equation.

> (a) Verified : The best line $b = 1 + 4t$ passes through $(2, 9)$.
> (b) Explained : $C + D\bar{t} = \bar{b}$ arises from the first normal equation.

C+2D=9 Athavadillan

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Exactly! You're spot on.
To break it down further:
When you're using **least squares** to fit a line (say,) to a set of data points , you're trying to minimize the total squared error between your predicted values and actual values.
The **normal equations** come from this minimization, and solving them gives you the best-fit parameters .
Now, when you **sum all the equations** (or look at the structure of the normal equations), it turns out that the resulting line must pass through the **centroid** of the data — the point:
(\bar{t}, \bar{b}) = \left(\frac{1}{n} \sum t_i, \frac{1}{n} \sum b_i\right)
So yes, the reason is always true for the least-squares line is because it's constructed to balance the errors, and the point is the **anchor point** that the line must pass through to achieve that balance.
Let me know if you'd like to see this shown with numbers or a small data example.

(12) $a = (1, \cdots 1) \rightarrow a^T a = m$

$a^T b = b_1 + \cdots b_m$

$\hat{x} = \frac{a^T b}{m}$ is the mean of the b's

b) $e = b - a\hat{x}$ $\|e\|^2 = (b_1 - \text{mean})^2 +$

... (bm − mean)

= variance = D^2

(c) $P = \frac{PP^T}{P^TP} = \frac{1}{27}\begin{bmatrix} 9 & 9 & 9 \\ \vdots & \vdots & \vdots \end{bmatrix} = \begin{matrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{matrix} \cdots$

$P = 3, 3, 3$

$b = (1, 2, 6)$

$e = 1-3,\ 2-3,\ 6-3 = -2, -1, 3$

(13) See ch gist. 4.3

(14)

(15)